

Analysis of pumped storage hydropower demand-supply gap in East China

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Abstract—This study proposes a four-step integrated framework to quantify the demand-supply gap of PSH (pumped storage hydropower) in East China (comprising Jiangsu, Zhejiang, Anhui provinces and Shanghai Municipality) for the target years 2030 and 2035. Firstly, theoretical demand is derived from hourly power-system simulations that co-optimize peak-shaving and renewable curtailment reduction using a peak-valley-difference model corrected by a 10–20 % renewable-integration coefficient. Secondly, supply-side capacity is assessed by linking project-specific construction timelines, environmental-approval status and equipment-delivery schedules to probabilistic commissioning curves for 23 approved PSH projects. Thirdly, provincial gaps are calculated by comparing demand and supply under 2030 and 2035 horizons and are stress-tested via sensitivity analysis on load-growth and renewable-deployment variations. Finally, inter-regional allocation scenarios are generated to prioritize surplus provinces (Zhejiang, Anhui) for deficit provinces (Jiangsu, Shanghai). Results show that East China will need 72.8 GW (Gigawatt) by 2035, leaving an aggregate gap of 29.715 GW (Shanghai 3.72 GW, Jiangsu 13.97 GW, Anhui 11.95 GW) while Zhejiang presents a 2.34 GW surplus. The framework provides actionable insights for PSH planning, cross-provincial trading and renewable integration. (*Abstract*)

Keywords- pumped storage hydropower; demand-supply gap; East China; renewable energy integration; power system (key words)

I. INTRODUCTION

The global energy transition has accelerated renewable energy penetration in power systems. As a critical flexibility resource, PSH plays a pivotal role in ensuring grid stability and facilitating renewable energy integration ^[1-2]. East China—an economically developed region with substantial energy demands—faces significant peak-shaving and renewable accommodation challenges due to large-scale renewable deployment. Consequently, scientifically analyzing PSH demand gaps is essential for optimizing regional power resource allocation and ensuring grid security.

Recent studies have extensively examined PSH demand forecasting, site optimization, and renewable-PSH coordination. International research emphasizes economic evaluation, technological advancements, and operational mechanisms in high-renewable systems ^[3-6]. Domestic studies focus more on site selection criteria, construction timelines, and regional grid compatibility ^[7-11]. However, existing work predominantly

concentrates on single-province assessments or static analyses, lacking cross-regional dynamic gap evaluations that account for actual project progress.

This study analyzes the "three provinces and one municipality" of East China by integrating multi-source data including power load forecasts, renewable energy plans, and PSH project schedules. We develop a demand-supply evaluation model that balances theoretical requirements with engineering feasibility, incorporating system peak-shaving needs and renewable accommodation requirements. Focusing on 2030 and 2035 planning horizons, we quantify provincial-level gaps through multidimensional assessment. Our findings reveal regional PSH resource imbalances and provide actionable insights for infrastructure planning and inter-provincial coordination, while supporting renewable integration and grid stability.

II. MATERIALS AND METHODS

This study employs a comprehensive analytical approach based on the current status and planned development of PSH in East China's "three provinces and one municipality" (Jiangsu, Zhejiang, Anhui, and Shanghai). First, we analyze the reasonable demand scale and spatial distribution of PSH plants across the region, incorporating data on construction progress and preparatory work for ongoing and planned projects. This allows us to evaluate the commissioning probability and capacity of PSH stations expected to be operational by 2035. Second, under the assumption that all planned PSH projects will be completed as scheduled, we assess the potential for additional PSH development in each province/municipality by 2035. Furthermore, through field investigations of construction timelines, project phases, and preparatory work status, we determine the practical feasibility of project completion and commissioning. Finally, based on provincial-level PSH demand analysis, we quantify the supply-demand gaps for 2030 and 2035, excluding the potential PSH project. Our methodology emphasizes the integration of actual project progress with planning objectives, providing multidimensional decision-making support for regional PSH development.

A. PSH Demand Forecasting

Considering peak-shaving requirements and renewable energy integration needs across provinces, we calculate the

theoretical PSH demand using the peak-valley difference method [12]:

$$P_{req} = P_{peak} - P_{min} - P_{other} \quad (1)$$

In the formula, P_{req} - Theoretical pumped-storage hydropower demand capacity, P_{peak} - the system peak load, P_{min} - the system minimum load, and P_{other} - the regulation capacity from other flexible resources (e.g., gas turbines, battery storage). To account for high renewable penetration, we apply a renewable integration adjustment:

$$P_{req} = k \times P_{RES} \quad (2)$$

In the formula, P_{RES} - the installed capacity of renewables, such as wind/PV (photovoltaic), k - the empirical coefficient (typically 10%-20%).

B. Supply-Demand Gap Analysis

For the target years 2030 and 2035, we compute provincial PSH gaps by comparing demand forecasts with supply capacity assessments:

$$Gap_t = P_{req,t} - \sum_{i \in S_t} P_i \quad (3)$$

In the formula, Gap_t - PSH capacity gap for year t (2030 or 2035), $P_{req,t}$ - PSH demand for year t , S_t - the set of PSH

plants expected to be operational by year t (2030 or 2035), P_i - Installed capacity of plant i . Sensitivity analysis is conducted on key variables (e.g., load growth rate, renewable capacity changes) to enhance prediction robustness.

III. RESULTS & DISCUSSION

A. Projected Commissioning Capacity by Target Year

As of the end of 2023, the operational PSH capacity in East China's "three provinces and one municipality" reached approximately 13.96 GW, representing the cumulative contribution of all plants that have successfully navigated construction, commissioning, and grid-connection phases. Concurrently, a detailed review of ongoing construction progress indicates that 23 PSH projects—each having secured full regulatory approval—are currently in various stages of development across the region, ranging from civil-works excavation to electro-mechanical installation. Assuming normal project advancement, all facilities are expected to be operational by 2035.

Taking into account the confirmed implementation status of existing assets, the firm schedules of under-construction schemes, and the realistic likelihood of planned projects reaching financial close, the total PSH capacity in East China is projected to reach 29.36 GW by 2030—a level that reflects both the incremental addition of new units and the scheduled retirement or refurbishment of legacy plants. Looking further ahead to 2035, the total is expected to climb to 43.09 GW, an expansion that is underpinned by the anticipated synchronization of the last tranche of approved projects. Fig 1 illustrates the anticipated commissioning scale across different provinces/municipalities.

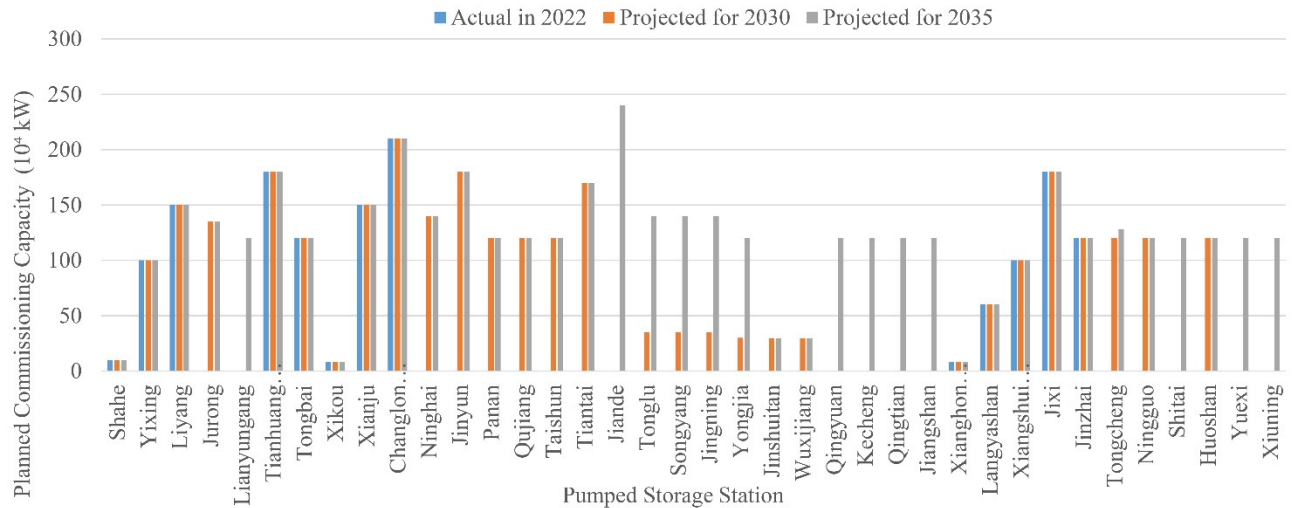


Fig. 1. Projected PSH commissioning capacity for 2030 and 2035

B. Spatial Analysis of PSH in the Three Provinces and One Municipality

1) Shanghai Power Grid

As of the end of 2022, the Shanghai Power Grid had a total available installed capacity of 30.87 GW, comprising 14.99 GW

of coal-fired power, 8.16 GW of gas-fired power, 2.57 GW of PSH, 1.07 GW of wind power, 1.95 GW of PV power, and 2.13 GW of biomass power. Imported electricity included 4.08 GW via 500 kV AC (kilovolt alternating current), 6.40 GW via 800 kV AC, and 6.22 GW through the East China back-to-back HVDC (High-Voltage Direct Current) link, summing to 16.70 GW.

In 2022, Shanghai's system-wide peak load reached 38.07 GW, an increase of 7.2 % year-on-year, while total electricity consumption was 174.6 TWh (terawatt-hour), a decrease of 0.2 % year-on-year. According to forecasts, the system peak load and electricity demand will reach 43.5 GW and 210 TWh, respectively, in 2030, and 46.5 GW and 222 TWh, respectively, in 2035. Power-structure optimisation studies indicate that the planned capacity of PSH for the Shanghai Power Grid is 5.0 GW in 2030, representing about 7.3 % of total installed capacity, and 6.3 GW in 2035, representing about 7.8 % of total installed capacity.

Shanghai possesses no indigenous PSH resources; therefore, all capacity must be provided by other provinces. At present, existing PSH plants in Zhejiang and Anhui already supply 2.58 GW to Shanghai—1.29 GW from Tianhuangping, Tongbai, and Xianju (Zhejiang) and 1.29 GW from Langyashan, Xiangshuijian, and Jixi (Anhui). An additional 2.42 GW of PSH capacity transmitted from other provinces will be required before 2030, and this figure rises to 3.72 GW before 2035.

2) Jiangsu Power Grid

As of the end of 2022, the Jiangsu Power Grid had a total installed capacity of 161.56 GW, comprising 78.92 GW of coal-fired power, 19.74 GW of gas-fired power, 6.61 GW of nuclear power, 0.05 GW of conventional hydropower, 2.60 GW of PSH, 22.54 GW of wind power, 25.09 GW of PV power, 2.88 GW of biomass power, and 0.30 GW of new-type energy storage. Imported electricity totalled 44.99 GW, of which 18.06 GW came from the Southwest, 21.00 GW from the "Three-North" regions, 3.22 GW via the East China Power Grid, and 2.71 GW via the Yangcheng Direct Line.

In 2022, Jiangsu's system-wide peak load reached 131.36 GW, an increase of 5.7 % year-on-year, while total electricity consumption was 740 TWh, an increase of 4.2 % year-on-year. According to forecasts, the system peak load and electricity demand will reach 176.0 GW and 930 TWh, respectively, in 2030, and 187.0 GW and 1,000 TWh, respectively, in 2035. Power-structure optimisation studies indicate that the planned capacity of PSH for the Jiangsu Power Grid is 17.0 GW in 2030, representing about 5.4 % of total installed capacity, and 21.0 GW in 2035, representing about 6.1 % of total installed capacity.

Jiangsu has limited indigenous PSH resources. At present, 2.60 GW are in operation (0.10 GW at Shahe, 1.00 GW at Yixing, and 1.50 GW at Liyang), and 2.55 GW are under construction (1.35 GW at Jurong and 1.20 GW at Lianyungang, expected to be commissioned in 2025 and 2031, respectively). All remaining demand must be met by imports. Existing PSH plants in Zhejiang and Anhui already supply 1.88 GW to Jiangsu—1.55 GW from Tianhuangping and Changlongshan (Zhejiang) and 0.33 GW from Jixi (Anhui). An additional 11.17 GW of PSH capacity transmitted from other provinces will be required before 2030, and this figure rises to 13.97 GW before 2035.

3) Zhejiang Power Grid

As of the end of 2022, the Zhejiang Power Grid had a total installed capacity of 117.93 GW, comprising 48.04 GW of coal-fired power, 12.62 GW of gas-fired power, 9.13 GW of nuclear power, 7.16 GW of conventional hydropower, 6.68 GW of PSH, 4.23 GW of wind power, 25.35 GW of PV power, 4.68 GW of biomass power, and 0.05 GW of new-type energy storage.

Imported electricity totalled 32.45 GW, including 2.34 GW via the 500 kV Three Gorges HVDC, 7.11 GW via AC UHV (ultra-high voltage), 20.0 GW via 800 kV HVDC, and 3.0 GW via 1100 kV HVDC.

In 2022, Zhejiang's system-wide peak load reached 108.6 GW, an increase of 6.7 % year-on-year, while total electricity consumption was 579.9 TWh, an increase of 6.6 % year-on-year. According to forecasts, the system peak load and electricity demand will reach 165.0 GW and 830 TWh, respectively, in 2030, and 191.0 GW and 952 TWh, respectively, in 2035. Power-structure optimisation studies indicate that the planned capacity of PSH for the Zhejiang Power Grid is 18.0 GW in 2030, representing about 7.2 % of total installed capacity, and 23.0 GW in 2035, representing about 7.7 % of total installed capacity.

Currently, 6.68 GW of PSH are in operation—1.80 GW at Tianhuangping, 1.20 GW at Tongbai, 1.50 GW at Xianju, 0.09 GW at Xikou and 2.10 GW at Changlongshan—of which 3.04 GW are exported. A further 21.7 GW have been approved and are under construction, with 10.45 GW expected to come online before 2030 and the full 21.7 GW before 2035. Consequently, Zhejiang faces a PSH demand gap of about 3.91 GW in 2030, whereas by 2035 its available PSH capacity is projected to exceed demand by 2.34 GW.

4) Anhui Power Grid

As of the end of 2022, the Anhui Power Grid had a total installed capacity of 92.20 GW, comprising 53.78 GW of coal-fired power, 1.54 GW of conventional hydropower, 4.68 GW of PSH, 5.90 GW of wind power, 21.54 GW of PV power, 2.42 GW of biomass power, and 2.33 GW of other sources. Inter-regional power exchange amounted to 7.75 GW (net export), consisting of 6.0 GW via 1100 kV HVDC, 1.40 GW via the East China back-to-back HVDC, 13.54 GW of "West-to-East Anhui" exports, and 1.61 GW of PSH exports.

In 2022, Anhui's system-wide peak load reached 108.6 GW, an increase of 6.7 % year-on-year, while total electricity consumption was 579.9 TWh, an increase of 6.6 % year-on-year. According to forecasts, the system peak load and electricity demand will reach 103.0 GW and 510 TWh, respectively, in 2030, and 135.0 GW and 650 TWh, respectively, in 2035. Power-structure optimisation studies indicate that the planned capacity of PSH for the Anhui Power Grid is 10.5 GW in 2030 and 22.5 GW in 2035.

Currently, 4.68 GW of PSH are in operation—0.08 GW at Xianghongdian, 0.60 GW at Langyashan, 1.00 GW at Xiangshuijian, 1.80 GW at Jixi and 1.20 GW at Jinzhai—of which 1.61 GW are exported. An additional 0.20 GW is received from Tianhuangping (Zhejiang). A further 7.28 GW have been approved and are under construction, with 3.60 GW expected to be commissioned before 2030 and the full 7.28 GW before 2035. Consequently, Anhui faces a PSH demand gap of about 3.63 GW in 2030, and the gap widens to about 11.95 GW by 2035.

C. PSH Demand-Supply Gap in East China

The East China region, comprising three provinces and one municipality, exhibits significant—and steadily growing—demand for PSH capacity. However, due to the prolonged construction cycles characteristic of PSH projects—typically

extending from the initial feasibility study through geotechnical investigations, environmental permitting, tunnelling, electro-mechanical installation and final grid-connection testing—planned developments are unlikely to meet the region's projected requirements within the designated timeframe. Through comprehensive analysis of preparatory work progress and construction timelines for ongoing PSH projects, we have evaluated the anticipated available capacity across provincial-level divisions during the target years. Our calculations indicate that by 2035, the total PSH demand across East China will reach approximately 72.8 GW. After accounting for the existing operational capacity of 13.96 GW and committed projects under construction totaling 29.125 GW, the region is projected to face an aggregate supply gap of 29.715 GW. This deficit demonstrates distinct regional variations, with Shanghai requiring an additional 3.72 GW, Jiangsu facing a 13.97 GW shortfall, Anhui needing 11.95 GW, while Zhejiang is expected to maintain a surplus of 2.34 GW.

The detailed spatial distribution of these capacity requirements is systematically presented in Tab 1, which provides a comprehensive overview of the PSH allocation scenarios for Shanghai, Jiangsu, and Zhejiang power grids by 2035, highlighting the critical need for inter-regional coordination to address these imbalances. Detailed breakdowns of the projected commissioned capacity—comprising both existing operational capacity and committed Projects Under Construction—for each grid are provided in Tab 2.

Tab. 1. PSH Capacity Allocation in East China Power Grids in 2035

Power Grid	Shanghai	Jiangsu	Zhejiang	Anhui
Planned Capacity	630	2100	2300	2250
Existing Operational Capacity	258	448	364	327
Committed Projects Under Construction	0	255	2170	728
Subtotal	258	703	2534	1055
Additional Required Capacity	372	1397	0	1195
Remarks	Requires external support	Requires external support	Surplus capacity	Self-sufficient

Unit: 10⁴ kW

Tab. 2. Projected Commissioned Capacity in East China Power Grids in 2035

Power Grid Name	Existing Operational Capacity (10 ⁴ kW)	Committed Projects Under Construction (10 ⁴ kW)
SH	ZJ-Tianhuangping 60 ZJ-Tongbai 50 ZJ-Xianju 19 AH-Langyashan 30 AH-Xiangshuijian 50 AH-Jixi 48.6	/
JS	JS-Yixing 100 JS-Shahe 10 JS-Liyang 150 ZJ-Tianhuangping 50 AH-Jixi 32.4 ZJ-Changlongshan 105	JS-Jurong 135 JS-Lianyungang 120

ZJ	ZJ-Tianhuangping 50 ZJ-Tongbai 70 ZJ-Xikou 8 ZJ-Xianju 131 ZJ-Changlongshan 105	ZJ-Ninghai 140 ZJ-Qujiang 120 ZJ-Taishun 120 ZJ-Jiande 240 ZJ-Songyang 140 ZJ-Yongjia 120 ZJ-Wuxijiang 29.8 ZJ-Kecheng 120 ZJ-Jiangshan 120	ZJ-Jinyun 180 ZJ-Panan 120 ZJ-Tiantai 170 ZJ-Tonglu 140 ZJ-Jingning 140 ZJ-Jinshuitan 29.7 ZJ-Qingyuan 120 ZJ-Qingtian 120
	AH-Langyashan 30 AH-Xiangshuijian 50 AH-Xianghongdian 8 AH-Jixi 99 AH-Jinzhai 120 ZJ-Tianhuangping 20	AH-Tongcheng 128 AH-Ningguo 120 AH-Huoshan 120 AH-Shitai 120 AH-Yuexi 120 AH-Xiuning 120	

Note: SH - Shanghai Power Grid, ZJ - Zhejiang Power Grid, JS - Jiangsu Power Grid, AH - Anhui Power Grid.

IV. CONCLUSIONS

Through comprehensive analysis and calculations:

(1) For the Shanghai Power Grid, by 2030, the system will require a total of 5.0 GW of pumped-storage hydropower to meet peak-shaving and renewable-integration targets; nevertheless, after deducting all firm contributions from existing plants and committed inter-provincial deliveries, a residual 2.42 GW shortfall remains. Extending the horizon to 2035, the planning guideline elevates the requirement to 6.3 GW, and the corresponding deficit widens to 3.72 GW

(2) For the Jiangsu Power Grid, although the 2030 target is set at 17.0 GW, the latest dispatch optimisation runs reveal an anticipated gap stands at 11.17 GW once planned retirements and derating factors are applied. Advancing to 2035, the strategic plan raises the target to 21.0 GW, yet the projected shortfall is still expected to reach 13.97 GW, reflecting Jiangsu's position as the region's foremost demand centre and the limited availability of indigenous PSH sites.

(3) For the Zhejiang Power Grid, the 2030 plan envisages 18.0 GW of installed PSH capacity; even accounting for accelerated commissioning schedules, a 3.91 GW deficit persists owing to aggressive renewable-curtailment mitigation targets. By 2035, however, the enlarged target of 23.0 GW is expected to yield a 2.34 GW surplus above demand, thereby positioning Zhejiang as a potential net exporter of PSH services to neighbouring grids.

(4) For the Anhui Power Grid, the 2030 requirement of 10.5 GW—derived from peak-demand-plus-reserve calculations—carries a 3.63 GW shortfall when compared against realistically achievable capacity additions. Extending the outlook to 2035, the expanded target of 22.5 GW will still leave an 11.95 GW gap, despite Anhui's abundant hydro-geological resources, due to competing land-use constraints and lengthy environmental permitting timelines.

Except for Zhejiang Province, which is expected to meet—indeed slightly exceed—its PSH demand by 2035, East China's "three provinces and one municipality" will face significant PSH capacity gaps in both 2030 and 2035, with Jiangsu exhibiting the largest deficit in absolute terms. Given the abundant PSH resources in Zhejiang and Anhui, any newly constructed PSH stations in East China should prioritize allocation to Jiangsu Province as the primary demand center, with Shanghai as the

secondary recipient to optimize regional resource distribution and minimize overall system-balancing costs.

While the present study provides a robust snapshot of the 2030 and 2035 PSH gaps under deterministic project timelines, several avenues merit further investigation. Firstly, embedding probabilistic construction-delay models and stochastic renewable-output scenarios will refine the uncertainty range around gap estimates. Also, coupling the PSH allocation framework with high-resolution power-market simulations could quantify the economic value of inter-provincial trading and dynamic pricing signals.

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